# ECE 447

## Lesson 13
## Angle Modulation, Part 3



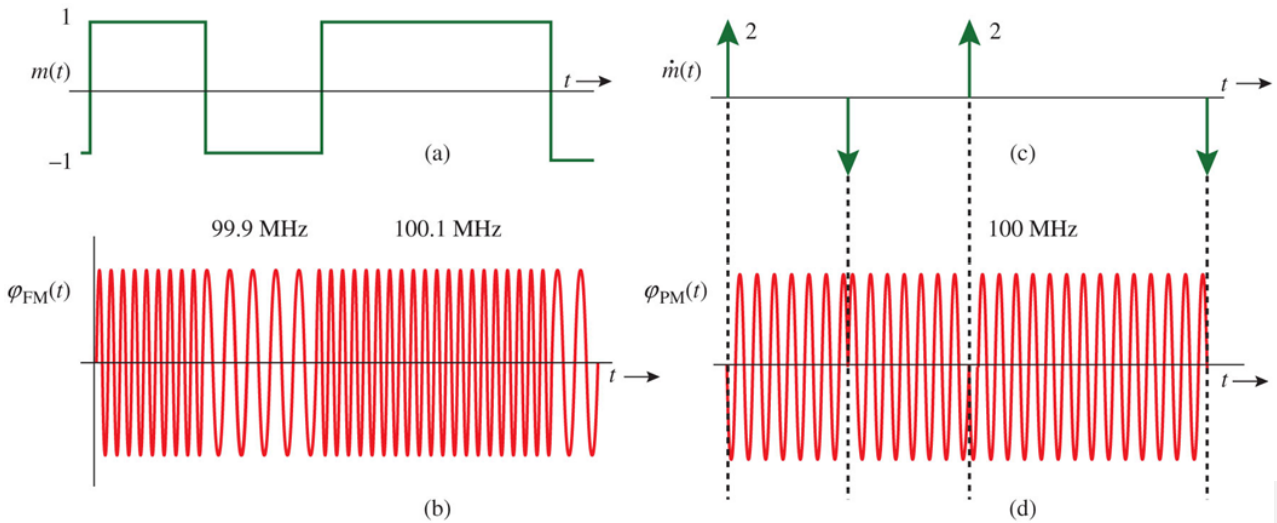
# Schedule and Admin

- Schedule
- Admin
  - HW3 due Tue, 15 Sep
  - **Lab 3 Assignment**. Due Lesson 16 (Tue, 22 Sep) by 2359 via Gradescope upload.
  - GR1 is Lesson 15 (Thu, 17 Sep). Review day is Lesson 14 (Tue, 15 Sep). Topics will primarily be from Chapter 4 of the textbook, though topics from Chapters 2-3 are fair game.

# REVIEW

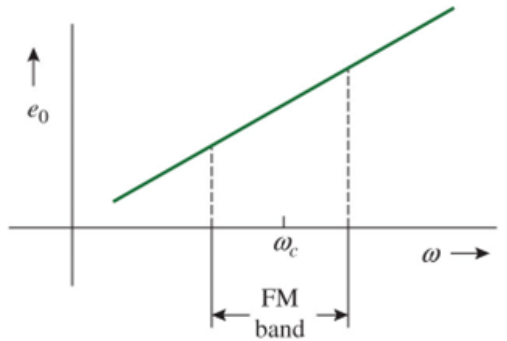
- If $m(t)$ has $m_p = 2V$ and a bandwidth of 50kHz, what is the deviation ratio $\beta$, assuming $k_f = 2,000\pi$? What is the FM signal bandwidth using Carson's Rule?

- Which attributes of $m(t)$ most influence the bandwidth of an FM signal? Of a PM signal?

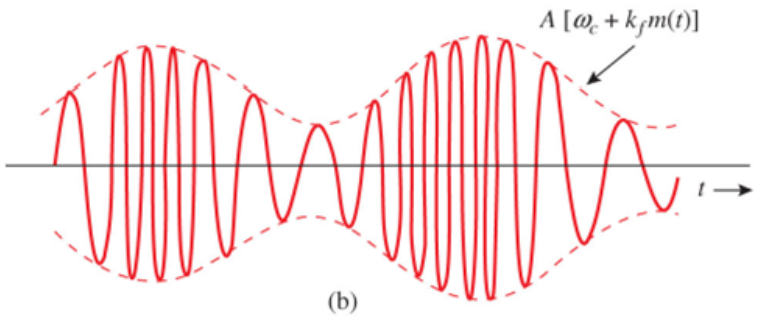


# FM Demodulation

- Basic idea: make FM look like AM and use an envelope detector
- Need output amplitude to vary as a function of input frequency
- Info of FM signal located in $\omega_i = \omega_c + k_f m(t)$
- Use ideal differentiator:
  $\dot{\varphi}_{FM}(t) = \frac{d}{dt}\left(A\cos[\omega_c t + k_f \int_{-\infty}^{t} m(\alpha)d\alpha]\right) = ?$

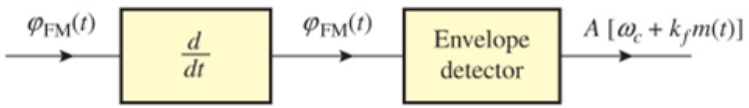


(a)



(b)

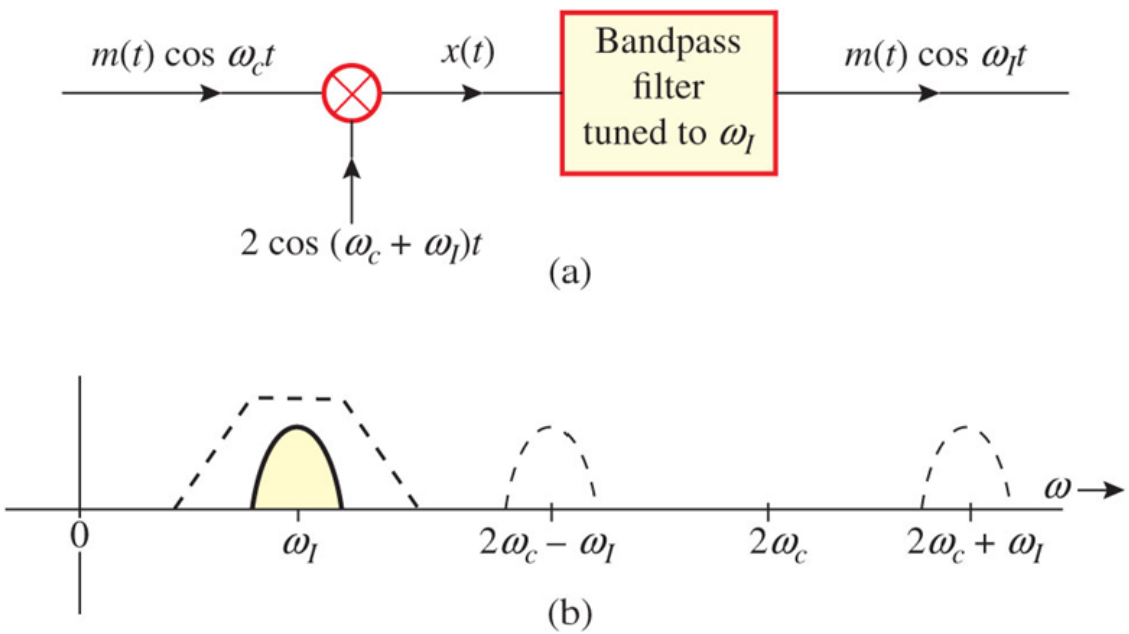
- Even though differentiator output is combo of FM and AM, envelope detector won't care if $\Delta f \ll f_c$ (typically true)



(c)

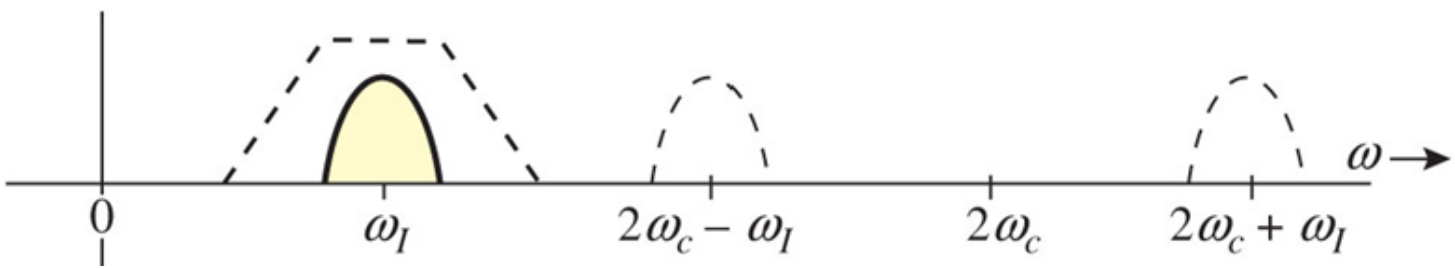
# Frequency Mixing, aka Heterodyne*

- Problem: generally filters can either be tunable without a sharp rolloff OR fixed with a sharp rolloff (i.e., high Q, or quality, factor)
- Solution: use both to get the best of both worlds
- Frequency mixer block diagram:



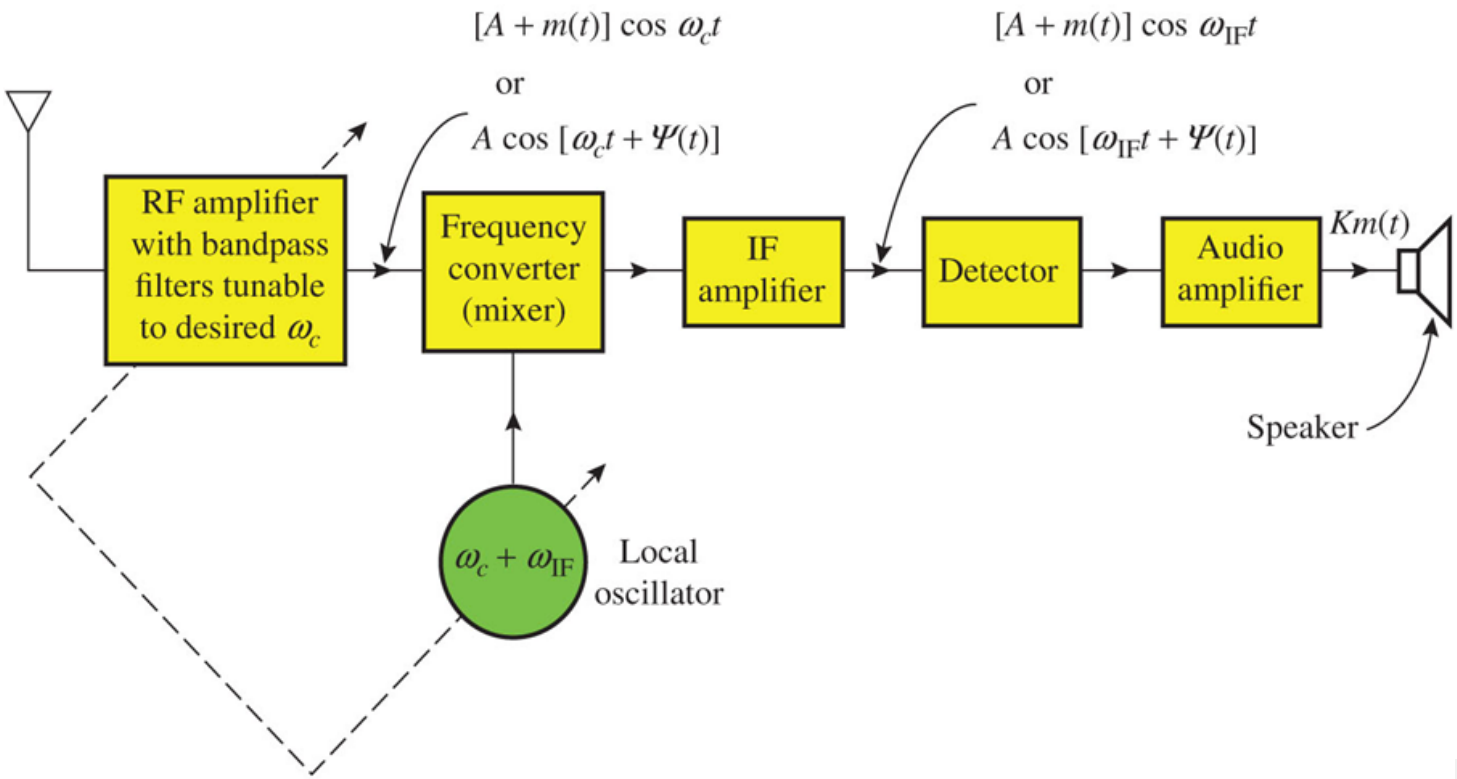

*The word "heterodyne" comes from Greek roots, combining hetero- ("other" or "different") and -dyne (from dýnamis, meaning "power" or "force"). Reginald Fessenden coined the term around 1901 to describe his principle of mixing two different radio frequencies to create an audible "beat" frequency in a wireless communication receiver.

# Frequency Mixing, aka Heterodyne*



# Superheterodyne Receiver

- Invented by Edwin Armstrong (see Section 4.8.4!)
- Widely used for broadcast receivers (AM, FM, TV)
- Uses a local oscillator (LO) with a frequency exactly $f_{IF}$ above $f_c$, or $f_{LO} = f_c + f_{IF}$
- Simultaneous tuning of both the LO and RF tunable filter (one knob does both)



# Image Stations

- **High-side LO injection**: the local oscillator is set above the desired carrier frequency:

$$f_{LO} = f_c + f_{IF}$$

- The mixer produces the intermediate frequency (IF): $f_{IF} = |f_{LO} - f_c|$
- Another station can also produce the same IF if its frequency is: $f_{\text{image}} = f_c + 2f_{IF}$
- This undesired station is called the **image station** – RF bandpass filter suppresses image frequency before mixing

**Example** – Given:

$$f_c = 1000 \text{ kHz}, \qquad f_{IF} = 455 \text{ kHz}$$

$$f_{LO} = 1000 + 455 = 1455 \text{ kHz}, \qquad f_{\text{image}} = f_c + 2f_{IF} = 1000 + 2(455) = 1910 \text{ kHz}$$

**Key Idea**

Both the desired station (1000 kHz) and the image station (1910 kHz) produce the same IF (455 kHz). The RF filter must reject the image station.

# FDM Steps

## Transmitter:

- Each signal uses a different carrier frequency
- Each signal may use a different kind of modulation (e.g., DSB-SC, FM, PM, SSB-USB, VSB-SC)
- The carrier frequencies are spaced to avoid overlap (also referred to as subcarriers)
- The composite signal may be considered a baseband signal and multiplied by an RF carrier before transmitting
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## Receiver:

- First demodulated by RF carrier to get composite baseband signal
- Bandpass-filtered to separate each modulated signals
- Each modulated signal is individually demodulated to get original baseband signal